# Omkar Sudhir Patil

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## Professional Summary

Researcher working at the intersection of controls, robotics, and AI to solve fundamental challenges in autonomous systems. Pioneered Lyapunov-based Deep Neural Networks (LbDNNs), solving a 25-year open problem by developing the first DNN-based controller with real-time adaptation and provable stability guarantees. Expertise spans adaptive control, multi-agent systems, and safety-critical applications, with research validated on real-world robotic platforms ranging from quadrotors to multi-robot coordination.

## Education

**University of Florida** *Aug 2019 – May 2023*
*Ph.D. in Mechanical Engineering* (GPA: 3.97/4.0)

- **Advisor:** Dr. Warren E. Dixon
- **Dissertation:** "Implicit and Deep Learning-Based Control Methods for Uncertain Nonlinear Systems"
- **Research Focus:** Lyapunov-based deep neural networks, adaptive control theory, nonlinear control, RISE controllers, multi-agent systems, concurrent learning, physics-informed neural networks
- **Key Achievement:** Solved 25-year open problem extending Lyapunov-derived adaptive online update laws from shallow networks to deep neural networks of arbitrary depth

**University of Florida** *Aug 2019 – Aug 2022*
*M.S. in Mechanical Engineering*

- **Specialization:** Nonlinear control systems, adaptive control, robust control

**Indian Institute of Technology Delhi** *Jul 2014 – Aug 2018*
*B.Tech in Production and Industrial Engineering*

- **Research Area:** Control systems and robotics applications

## Professional Experience

**Research Scientist** *Oct 2025 – Present*
*University of Florida, Department of Mechanical & Aerospace Engineering*

- **Advanced Control Systems:** Leading research on next-generation mathematical frameworks for modular and compositional control design, enabling scalable solutions for complex autonomous systems
- **Energy-Based Methods:** Developing innovative control approaches leveraging energy principles and geometric structures for improved stability and efficiency in dynamic systems
- **Ongoing Applications:** Advancing adaptive control solutions for aerospace systems, autonomous spacecraft servicing, and multi-agent robotics with guaranteed performance

**Postdoctoral Research Associate** *May 2023 – Oct 2025*
*University of Florida, Department of Mechanical & Aerospace Engineering*

- **Research Funding:** Secured $2M+ in research funding from ONR, AFOSR, and AFRL for adaptive control and deep learning projects
- **Breakthrough Innovations:** Pioneered Lyapunov-based deep neural network (Lb-DNN) controllers with online weight adaptation, enabling real-time learning without pre-training requirements
- **Industry Applications:** Developed adaptive control solutions for quadrotor control, spacecraft servicing, and multi-agent robotics with stability guarantees

- **Publication Impact:** Published 20 papers in premier venues including IEEE TAC, IEEE T-NN, ACC, and CDC with 300+ citations
- **Mentorship:** Supervised 15 graduate students and postdocs, fostering collaborative research environment
- **Software Development:** Created MATLAB and Python packages for Lb-DNN controllers and projection algorithms, deployed on robotic platforms at UF Autonomy Park

**Graduate Research Assistant** *Aug 2019 – May 2023*
*University of Florida, Nonlinear Controls and Robotics Group*

- **Theoretical Breakthroughs:** Solved 17-year open question on exponential stability of RISE controllers, previously thought to achieve only asymptotic convergence
- **Algorithm Development:** Developed composite adaptive laws for deep neural networks using concurrent learning, achieving parameter convergence without persistent excitation
- **Multi-disciplinary Research:** Advanced control barrier functions, physics-informed neural networks, and stochastic control theory
- **Experimental Validation:** Implemented controllers on quadrotors, robotic manipulators, and multi-agent systems with real-time performance verification

**Project Associate** *May 2018 – Apr 2019*
*Indian Institute of Technology Delhi, Department of Electrical Engineering*

- **Aerial Robotics:** Designed and implemented hierarchical control systems for quadrotor aerial manipulators
- **Nonlinear Control:** Developed singularity-free control algorithms using barrier Lyapunov functions and saturation techniques
- **System Integration:** Programmed flight control systems and derived complete Lagrangian dynamics for aerial manipulation platforms

## Key Research Contributions & Impact

### Deep Learning & Control Theory Integration

- **Pioneering Framework:** First to extend Lyapunov-based adaptive control to deep neural networks of arbitrary depth, solving 25-year theoretical gap
- **Composite Adaptation:** Developed first composite adaptive laws for Lb-DNNs using concurrent learning, achieving parameter convergence guarantees
- **Real-time Learning:** Created online adaptation algorithms requiring no pre-training, enabling deployment in unknown dynamic environments

### Fundamental Control Theory Advances

- **RISE Stability:** Proved exponential stability of RISE controllers, resolving 17-year open question in robust control
- **Stochastic Control:** Extended Lb-DNN framework to stochastic nonlinear systems with unknown drift and diffusion processes
- **Safety-Critical Systems:** Advanced adaptive control barrier functions using deep neural networks for real-time safety guarantee

### Multi-Agent & Aerospace Applications

- **Spacecraft Control:** Developed distributed controllers for collaborative spacecraft servicing under partial state feedback
- **Target Tracking:** Created exponential convergence algorithms for heterogeneous multi-agent systems with limited communication
- **Herding Control:** Designed adaptive indirect herding strategies for multiple target agents with unknown interaction dynamics

## Technical Skills

**Control Theory & Mathematics:** Lyapunov stability analysis, adaptive control, nonlinear control, robust control, optimal control, stochastic control, multi-agent systems, differential geometry, functional analysis, convex optimization

**Machine Learning & AI:** Deep neural networks, physics-informed neural networks, reinforcement learning, concurrent learning, online learning algorithms, ResNet architectures, LSTM networks, graph neural networks

**Programming & Software:** Python, C++, ROS, Git, MATLAB/Simulink, TensorFlow, PyTorch, control system toolboxes, numerical optimization solvers

**Experimental Platforms:** Quadrotor UAVs, robotic manipulators, multi-agent testbeds, real-time control systems, embedded systems, hardware-in-the-loop simulation

## Interactive Research Demonstrations

- **LbDNN Quadrotor Control:** 3D interactive simulation demonstrating real-time neural network adaptation with feedback loss
- **CBF Lab:** Educational platform for Control Barrier Functions with safety-critical control demonstrations
- **Physics-Informed Neural Networks:** Interactive demos for scientific machine learning with automatic differentiation
- **Neural ODE Visualizer:** Real-time continuous-time deep learning with dynamic system modeling
- **SLAM Tutorial:** Interactive robotics localization and mapping educational tool

## Selected Publications

### Books

[1] **O. Patil**, E. Griffis, and W. E. Dixon, *Lyapunov-Based Deep Learning for Autonomy.* In development with Springer (anticipated completion May 2026).

### Journal Papers

[2] **O. Patil**, R. Sun, S. Bhasin, and W. E. Dixon, "Adaptive control of time-varying parameter systems with asymptotic tracking," *IEEE Trans. Autom. Control*, vol. 67, no. 9, pp. 4809–4815, 2022.

[3] **O. Patil**, D. Le, M. Greene, and W. E. Dixon, "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network," *IEEE Control Syst Lett.*, vol. 6, pp. 1855–1860, 2022.

[4] **O. Patil**, A. Isaly, B. Xian, and W. E. Dixon, "Exponential stability with RISE controllers," *IEEE Control Syst. Lett.*, vol. 6, pp. 1592–1597, 2022.

[5] **O. Patil**, R. Kamalapurkar, and W. Dixon, "A saturated RISE controller with exponential stability guarantees," *IEEE Trans. Autom. Control*, 2025.

[6] **O. Patil**, D. Le, E. Griffis, and W. Dixon, "Lyapunov-based deep residual neural network (ResNet) adaptive control," *IEEE Access*, 2025.

[7] **O. Patil**, B. C. Fallin, C. F. Nino, R. G. Hart, and W. E. Dixon, "Bounds on deep neural network partial derivatives with respect to parameters," *arXiv preprint arXiv:2503.21007*, 2025.

[8] A. Isaly, **O. Patil**, H. Sweatland, R. Sanfelice, and W. E. Dixon, "Adaptive safety with a RISE-based disturbance observer," *IEEE Trans. Autom. Control*, vol. 69, no. 7, pp. 4883–4890, 2024.

[9] E. Griffis, **O. Patil**, Z. Bell, and W. Dixon, "Lyapunov-based long short-term memory (Lb-LSTM) neural network-based control," *IEEE Control Syst. Lett.*, vol. 7, pp. 2976–2981, 2023.

[10] C. Nino, **O. Patil**, and W. Dixon, "Second-order heterogeneous multi-agent target tracking without relative velocities," *IEEE Control Systems Letters*, vol. 7, pp. 3663–3668, 2023.

[11] E. Griffis, **O. Patil**, R. Hart, and W. Dixon, "Lyapunov-based long short-term memory (Lb-LSTM) neural network-based adaptive observer," *IEEE Control Syst. Lett.*, vol. 8, pp. 97–102, 2024.

[12] R. Hart, E. Griffis, **O. Patil**, and W. Dixon, "Lyapunov-based physics-informed long short-term memory (LSTM) neural network-based adaptive control," *IEEE Control Syst. Lett.*, vol. 8, pp. 13–18, 2024.

[13] D. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for deep neural network-based adaptive control of unknown nonlinear systems," *IEEE Trans. Neural Netw. Learn. Syst.*, 2025.

[14] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," *The Journal of the Astronautical Sciences*, 2025.

[15] C. Nino, **O. Patil**, J. Insinger, M. Eisman, and W. E. Dixon, "Online resnet-based adaptive control for nonlinear target tracking," *IEEE Control Syst. Lett.*, 2025.

[16] C. Nino, **O. Patil**, S. Edwards, and W. E. Dixon, "Distributed RISE-based control for exponential heterogeneous multi-agent target tracking of second-order nonlinear systems," *IEEE Control Syst. Lett.*, 2025.

[17] W. Makumi, **O. Patil**, and W. Dixon, "Lyapunov-based adaptive deep learning for approximate dynamic programming," *Automatica*, vol. 180, p. 112 462, 2025.

[18] E. Griffis, **O. Patil**, W. Makumi, and W. Dixon, "Adaptive output feedback control using lyapunov-based deep recurrent neural networks (Lb-DRNNs)," *IEEE Trans. Autom. Control*, 2025.

[19] R. Hart, **O. Patil**, Z. Bell, and W. Dixon, "Concurrent learning for system identification and control using lyapunov-based deep neural networks," *IEEE Control Syst. Lett.*, vol. 9, pp. 2957–2962, 2025.

## Conference Papers

[20] **O. Patil**, D. M. Le, E. Griffis, and W. E. Dixon, "Deep residual neural network (ResNet)-based adaptive control: A Lyapunov-based approach," in *Proc. IEEE Conf. Decis. Control*, 2022.

[21] **O. Patil**, K. Stubbs, P. Amy, and W. E. Dixon, "Exponential stability with RISE controllers for uncertain nonlinear systems with unknown time-varying state delays," in *Proc. IEEE Conf. Decis. Control*, 2022.

[22] **O. Patil** and S. Bhasin, "Lyapunov based hierarchical trajectory control of an autonomous ground vehicle subjected to slip," in *IFAC World Congr.*, 2020.

[23] D. M. Le, **O. Patil**, C. Nino, and W. E. Dixon, "Accelerated gradient approach for neural network-based adaptive control of nonlinear systems," in *Proc. IEEE Conf. Decis. Control*, 2022.

[24] D. M. Le, **O. Patil**, P. Amy, and W. E. Dixon, "Integral concurrent learning-based accelerated gradient adaptive control of uncertain Euler-Lagrange systems," in *Proc. Am. Control Conf.*, Jun. 2022.

[25] A. Isaly, **O. Patil**, R. Sanfelice, and W. E. Dixon, "Adaptive safety with multiple barrier functions using integral concurrent learning," in *Proc. Am. Control Conf.*, 2021, pp. 3719–3724.

[26] R. Dasgupta, S. B. Roy, **O. Patil**, and S. Bhasin, "A singularity-free hierarchical nonlinear quad-rotorcraft control using saturation and barrier Lyapunov function," in *Proc. Am. Control Conf.*, 2019.

[27] E. Griffis, **O. Patil**, W. Makumi, and W. E. Dixon, "Deep recurrent neural network-based observer for uncertain nonlinear systems," in *IFAC World Congr.*, 2023.

[28] R. Hart, **O. Patil**, E. Griffis, and W. E. Dixon, "Lyapunov-based deep physics-informed neural network-based adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2023.

[29] C. Nino, **O. Patil**, J. Philor, Z. Bell, and W. Dixon, "Deep adaptive indirect herding of multiple target agents with unknown interaction dynamics," in *Proc. IEEE Conf. Decis. Control*, 2023.

[30] C. F. Nino, **O. Patil**, S. C. Edwards, Z. I. Bell, and W. E. Dixon, "Distributed target tracking under partial feedback using Lyapunov-based deep neural networks," in *Proc. Am. Control Conf.*, 2025.

[31] C. F. Nino, **O. Patil**, C. D. Petersen, S. Phillips, and W. E. Dixon, "Collaborative spacecraft servicing under partial feedback using Lyapunov-based deep neural networks," in *Proc. AAS/AIAA Sp. Flight Mech. Meet.*, 2025.

[32] W. Wu, **O. Patil**, H. Sweatland, and W. Dixon, "Control contraction metric-based deep neural network adaptive control," in *Proc. IEEE Conf. Decis. Control*, 2025.

[33] S. Kolhe, **O. Patil**, J. Fu, and W. Dixon, "Integral concurrent learning control barrier functions for signal temporal logic tasks under unknown dynamics," in *Proc. IEEE Conf. Decis. Control*, 2025.

## Under Review


[34] **O. Patil**, E. Griffis, W. Makumi, and W. E. Dixon, "Simultaneous online system identification and control using composite adaptive Lyapunov-based deep neural networks," *IEEE Trans. Autom. Control*, 2025, under review, Preprint: https://arxiv.org/abs/2311.13056.

[35] H. Sweatland, **O. Patil**, and W. Dixon, "Adaptive deep neural network-based control barrier functions," *IEEE Trans. Autom. Control*, under review, Preprint: https://arxiv.org/abs/2406.14430.

[36] S. Akbari, E. J. Griffis, **O. Patil**, and W. E. Dixon, "Lyapunov-based dropout deep neural network (Lb-DDNN) controller," *Automatica*, under review, Preprint: https://arxiv.org/abs/2310.19938.

[37] B. Fallin, C. Nino, **O. Patil**, and W. Dixon, "Lyapunov-based graph neural networks for adaptive control of multi-agent systems," *IEEE Trans. Autom. Control*, under review, Preprint: https://arxiv.org/abs/2503.15360.

[38] S. Akbari, C. Nino, **O. Patil**, and W. Dixon, "Lyapunov-based deep neural networks for adaptive control of stochastic nonlinear systems," *IEEE Trans. Autom. Control*, under review, Preprint: https://arxiv.org/abs/2412.21095.

[39] R. Hart, **O. Patil**, Z. Bell, and W. Dixon, "Lyapunov-based concurrent learning for dnn parameter identifiability without persistent excitation," *IEEE Trans. Autom. Control*, under review, Preprint: https://arxiv.org/abs/2505.10678.

[40] S. Akbari, **O. Patil**, and W. Dixon, "LyLA-therm: Lyapunov-based Langevin adaptive thermodynamic neural network controller," *IEEE Trans. Autom. Control*, under review, Preprint: https://arxiv.org/abs/2508.14989.

[41] Y. Wang, **O. Patil**, and W. Dixon, "Riemannian lyapunov optimizer: A unified framework for optimization," *Proceedings of the International Conference on Machine Learning (ICML)*, under review, Preprint: https://arxiv.org/abs/2601.22284.


## Awards

- **Graduate Student Research Award**, 2023, for outstanding research as a graduate student in the Department of Mechanical and Aerospace Engineering, University of Florida
- **BOSS Award**, 2018, for the best hardcore experimental project in Mechanical Engineering discipline, IIT Delhi

## Invited Talks

- "Lyapunov-based deep learning for safe and reliable autonomy", *Research Scientist Seminar, University of Florida*, Sep. 2025
- "Lyapunov-based deep learning for safe and reliable autonomy", *Graduate Fluids Seminar, University of Florida*, Oct. 2025
- "Simultaneous System Identification and Control using Composite Adaptive LbDNNs", *AFOSR Program Review, Air Force Office of Scientific Research, Washington DC*, April 2025
- "Composite Adaptive Lyapunov-based Deep Neural Network Control", *AFOSR COE Program Review, Duke University*, Dec. 2023
- "Deep Residual Neural Network (ResNet)-based Adaptive Control: A Lyapunov-based Approach", *AFOSR COE Program Review, University of Florida*, April. 2023
- "Lyapunov-derived control and adaptive update laws for inner and outer layer weights of a deep neural network", *Resilient and Autonomous Systems Lab (RASLab), Florida State University*, Dec. 2021

## Teaching Experience

- Teaching Assistant (UF), Vibrations, Spring 2020

## Peer Review Service

- Nature Communications
- IEEE Transactions on Automatic Control
- Automatica
- IEEE Control Systems Letters
- IEEE Transactions on Control Systems Technology
- IEEE Transactions on Signal Processing
- IEEE/ASME Transactions on Mechatronics
- IEEE Transactions on Neural Networks and Learning Systems
- International Journal of Adaptive Control and Signal Processing
- Advances on Neural Information Processing Systems (NeurIPS)
- International Conference on Learning and Representations (ICLR)
- International Conference on Machine Learning (ICML)
- AAAI Conference on Artificial Intelligence (AAAI)
- IEEE Conference on Decision and Control
- American Control Conference
- IFAC World Congress